Differentiation — Chain Rule Skill Ladder

NCEA Level 3 Maths, AS91578 · one tiny skill at a time · do not skip ahead

Name: _____ Date: _____

How to use this sheet: Do Chunk 0 completely (read the explanation, do all 10 questions, check them against the answer sheet) before even looking at Chunk 1. Each chunk only needs the skill from the chunk(s) before it — nothing more. If you get more than 2 wrong in a chunk, stop and redo that chunk's explanation before moving on.

The ladder: Chunk 0 special angle values → Chunk 1 basic derivative facts → Chunk 2 derivative of a straight line expression → Chunk 3 spot outer/inner (easy case) → Chunk 4 chain rule (easy case) → Chunk 5 spot outer/inner (harder case) → Chunk 6 derivative of the inner piece alone → Chunk 7 chain rule (harder case) → Chunk 8 chain rule twice in a row. No product rule yet — that's the next sheet.

Chunk 0 — Special angle values (sin, cos, tan, sec)

The rule: sec x means 1/cos x. So $\sec^2 x$ means 1/(cos x), all squared. You need these five angles (0, π/6, π/4, π/3, π/2) instantly — not re-derived from a triangle every time — because a chain-rule question will bury one of these values halfway through a much bigger problem.

angle	0	π/6	π/4	π/3	π/2
sin	0	1/2	√2/2	√3/2	1
cos	1	√3/2	√2/2	1/2	0
tan	0	√3/3	1	√3	undefined
sec	1	2√3/3	√2	2	undefined
$\sec^2$	1	4/3	2	4	undefined

WORKED EXAMPLE

Evaluate $\sec^2(0)$.

Step 1: cos(0) = 1.

Step 2: sec(0) = 1/cos(0) = 1/1 = 1.

Step 3: $\sec^2(0) = (\sec 0)^2 = 1^2 = 1$.

Answer: 1

Your turn — evaluate each. Use the table above until these are automatic.

1. sin(0)

Answer:

2. cos(π/4)

Answer:

3. tan(π/3)

Answer:

4. sec(0)

Answer:

5. $\sec^2(0)$

Answer:

6. $\cos(\pi/6)$

Answer:

7. $\sin(\pi/2)$

Answer:

8. $\tan(\pi/4)$

Answer:

9. $\sec^2(\pi/4)$

Answer:

10. $\sec^2(\pi/3)$

Answer:

Chunk 1 — Basic derivative facts (no chain rule yet)

The rules to know cold:

$d/dx[x^n] = n \cdot x^{n-1}$ · $d/dx[e^x] = e^x$ · $d/dx[\ln x] = 1/x$ · $d/dx[\sin x] = \cos x$ · $d/dx[\cos x] = -\sin x$ · $d/dx[\tan x] = \sec^2 x$ · $d/dx[\text{constant}] = 0$

WORKED EXAMPLE

Differentiate $y = x^4$.

Power rule, n=4: $dy/dx = 4x^{4-1}$

Answer: $dy/dx = 4x^3$

Your turn — differentiate each. These are all single, plain functions — nothing is "inside" anything else yet.

1. $y = x^5$

dy/dx =

2. $y = e^x$

dy/dx =

3. $y = \ln x$

dy/dx =

4. $y = \sin x$

dy/dx =

5. $y = \cos x$

dy/dx =

6. $y = \tan x$

dy/dx =

7. $y = 7$

dy/dx =

8. $y = x$

dy/dx =

9. $y = \sqrt{x}$ $(= x^{1/2})$

dy/dx =

10. $y = 1/x$ $(= x^{-1})$

dy/dx =

Chunk 2 — Derivative of a straight-line expression (ax+b)

The rule: the derivative of ax+b is always just a — the x-coefficient, and nothing else. The constant b always disappears completely (its derivative is 0). This is the building block you'll multiply by in the next chunk, so it needs to be instant.

WORKED EXAMPLE

Differentiate $y = 6x - 11$.

Answer: dy/dx = 6

Your turn — just the x-coefficient.

1. $3x+5$

derivative =

2. $7x-2$

derivative =

3. $-4x+1$

derivative =

4. $x/2+3$

derivative =

5. $6-2x$

derivative =

6. $5x$

derivative =

7. $9x+4$

derivative =

8. $-x+7$

derivative =

9. $2x-11$

derivative =

10. $10x-1$

derivative =

Chunk 3 — Spot the outer and inner function (easy case)

The rule: ask yourself — "if I substituted a number for x, what's the LAST thing I'd do?" That last operation is the OUTER function. Everything it's being done TO is the INNER function. No differentiating in this chunk — just naming the two layers.

WORKED EXAMPLE

For $y = e^{7x-2}$, name the outer and inner function.

Last operation: raising e to a power. That power is the inner function.

Answer: Outer = e^(box). Inner = 7x−2.

Your turn — write the outer function (using "box" for the inner part) and the inner function. Don't differentiate anything yet.

1. e^{3x+1}

Outer: Inner:

2. $\sin(2x-5)$

Outer: Inner:

3. $(4x+7)^3$

Outer: Inner:

4. $\ln(5x+2)$

Outer: Inner:

5. $\cos(6x)$

Outer: Inner:

6. $(x+9)^5$

Outer: Inner:

7. e^{-2x}

Outer: Inner:

8. $\tan(3x+1)$

Outer: Inner:

9. $\sqrt{(2x+7)}$

Outer: Inner:

10. $(10-x)^4$

Outer: Inner:

Chunk 4 — Chain rule, full computation (easy case: straight-line inner)

The 3-step method:
1. Apply the normal derivative rule to the OUTER function, leaving the inner exactly as it is.
2. Multiply by the derivative of the inner function (from Chunk 2 — just a number here).
3. Tidy up.

WORKED EXAMPLE

Differentiate $y = \sin(5x+3)$.

Step 1: outer rule on sin(box) → $\cos(5x+3)$, inner untouched.

Step 2: inner derivative = 5.

Step 3: multiply.

Answer: $dy/dx = 5\cos(5x+3)$

Your turn — full derivative this time, using the 3-step method.

1. $y = e^{5x+2}$

2. $y = \sin(4x-1)$

3. $y = (3x+1)^4$

4. $y = \ln(2x+7)$

5. $y = \cos(5x)$

6. $y = (x-6)^3$

7. $y = e^{-3x}$

8. $y = \tan(2x+5)$

9. $y = \sqrt{(3x+4)}$

10. $y = (8-2x)^5$

Chunk 5 — Spot the outer and inner function (harder case)

Same skill as Chunk 3 — same question ("what's the last operation?") — but now the inner function is more than just ax+b. Still no differentiating, just naming the two layers.

WORKED EXAMPLE

For $y = \cos(x^3)$, name the outer and inner function.

Answer: Outer = cos(box). Inner = x^3.

Your turn — name the outer and inner function. Don't differentiate.

1. e^{x^2}

Outer: Inner:

2. $\sin(x^3)$

Outer: Inner:

3. $(x^2+1)^5$

Outer: Inner:

4. $\ln(x^2+3)$

Outer: Inner:

5. $\cos(\sqrt{x})$

Outer: Inner:

6. $(1/x + 1)^3$

Outer: Inner:

7. $\tan(x^2)$

Outer: Inner:

8. e^{4-x^2}

Outer: Inner:

9. $\ln(3x^3)$

Outer: Inner:

10. $\sqrt{(x^2+1)}$

Outer: Inner:

Chunk 6 — Differentiate the inner piece on its own

The rule: before attempting the full chain rule, make sure you can differentiate whatever is INSIDE, by itself, as its own mini-question. This is just Chunk 1's power rule again, applied to the kind of expressions that show up as "inner functions."

WORKED EXAMPLE

Differentiate x^2+1 (on its own, as if it were the whole question).

Answer: 2x

Your turn — just differentiate each expression on its own, nothing wrapped around it.

1. x^2

derivative =

2. x^3

derivative =

3. x^2+1

derivative =

4. x^2+3

derivative =

5. $\sqrt{x}$

derivative =

6. $1/x + 1$

derivative =

7. $4-x^2$

derivative =

8. $3x^3$

derivative =

9. x^2-5x

derivative =

10. $1/x$

derivative =

Chunk 7 — Chain rule, full computation (harder case)

Same 3-step method as Chunk 4 — outer rule first, leave inner untouched, then multiply by the derivative of the inner — except now step 2 uses your Chunk 6 skill, because the inner's derivative is a function of x, not just a plain number.

WORKED EXAMPLE

Differentiate $y = (x^2+1)^3$.

Step 1: outer rule on $(\text{box})^3 \rightarrow 3(x^2+1)^2$, inner untouched.

Step 2: inner derivative (Chunk 6) = $2x$.

Step 3: multiply: $3(x^2+1)^2 \times 2x$.

Answer: $dy/dx = 6x(x^2+1)^2$

Your turn — full derivative, 3-step method.

1. $y = e^{x^2}$

2. $y = \sin(x^3)$

3. $y = (x^2+1)^5$

4. $y = \ln(x^2+3)$

5. $y = \cos(\sqrt{x})$

6. $y = \tan(x^2)$

7. $y = e^{4-x^2}$

8. $y = \ln(3x^3)$

9. $y = \sqrt{(x^2+1)}$

10. $y = (x^2-5x)^4$

Chunk 8 — Chain rule twice in a row (three layers)

The rule: when there are THREE layers instead of two, peel off the outside first exactly as before — but now the "inner function" you have to multiply by is itself a chain-rule problem, so you repeat the whole 3-step method a second time, on just that piece, before you're done.

WORKED EXAMPLE

Differentiate $y = e^{\cos 2x}$.

Step 1 (outer = e^(box), inner = cos2x): $dy/dx = e^{\cos 2x} \times$ (derivative of cos2x).

Step 2 (that derivative is its OWN chain-rule job — outer cos(box), inner 2x): derivative of cos2x = $-\sin(2x) \times 2 = -2\sin 2x$.

Step 3: combine the two results.

Answer: $dy/dx = -2\sin(2x) \cdot e^{\cos 2x}$

Your turn — every one of these needs the chain rule applied twice. Go outer, then middle, then inner, multiplying as you go.

1. $y = e^{\sin 2x}$

2. $y = \sin(e^{3x})$

3. $y = \ln(\cos 5x)$

4. $y = (\sin 3x)^4$

5. $y = \cos(\ln 2x)$

6. $y = \sqrt{(\sin 4x)}$

7. $y = (e^{2x}+1)^3$

8. $y = \ln(\sin 4x)$

9. $y = \sin((2x+1)^2)$

10. $y = e^{\cos 3x}$